DATE	28-02-2026
TEAM ID	LTVIP2026TMID*****
PROJECT NAME	HEMATOVISION: ADVANCED BLOOD CELL CLASSIFICATION USING TRANSFER LEARNING
MAXIMUM MARKS	4 MARKS

4.1 PROBLEM SOLUTION FIT

Problem–Solution Fit explains how the proposed system, HematoVision: Advanced Blood Cell Classification Using Transfer Learning, effectively addresses the identified challenges in manual blood cell analysis and medical diagnostics.

Identified Problems:

Manual blood cell classification is time-consuming and labor-intensive.
Results depend heavily on the expertise of laboratory professionals.
High possibility of human error and inconsistent interpretation.
Increasing patient data leads to delays in diagnosis.
Developing deep learning models from scratch requires large datasets and long training time.

Proposed Solution:

HematoVision provides an automated blood cell classification system using transfer learning with pre-trained Convolutional Neural Networks (CNNs). The system processes microscopic blood cell images, performs preprocessing, and accurately classifies different cell types with high reliability.

How the Solution Fits the Problem:

- Reduces Time Consumption: Automation significantly speeds up the classification process compared to manual examination.
- Improves Accuracy: Transfer learning enhances model performance using knowledge from pre-trained models.
- Ensures Consistency: The AI model provides standardized results, reducing variability in diagnosis.
- Minimizes Training Time: Using pre-trained CNN models reduces computational cost and development time.
- Supports Medical Professionals: Acts as a decision-support tool to assist doctors and laboratory technicians.